SUPERVISED BY

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sion process

n decisions rely solely on academic ecific aptitude and potential.

: Each department uses different across the university's admission

ed tests and manual grading are slow, le for thousands of applicants.

lment data on incoming cohort quality, ademic support.

Background



ersity-wide digital admission test
ograms at DIU.

sess Math, Logic, IT, English, and
-minute session.

bjective scoring with eligibility
nual grading errors.

n-wise analytics and merit lists to
decisions.

i-cheating system ensuring fair

Objective



comes

tion

esponsive SPA with candidate portal, engine.

stant results with section-wise classification.

ve charts for pass rates, section hort profiling.

nked candidate list with eligibility

able individual performance reports

rds.

Outcomes

in Panel

d with live
Question
D + bulk
rt · Exam
ion
timing,
s) ·

ent · Merit
nking ·
with bar,
he charts

🔒 Anti-Cheat System

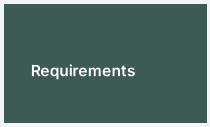
Tab-switch detection · Fullscreen enforcement · Copy/paste/cut blocking · Right-click disabled · Keyboard shortcut blocking · DevTools detection · Full audit trail with per-question time

tracking

Features

quirements

07	Question Bank Management (CRUD)
08	Bulk Question Import (CSV)
09	Exam Configuration Panel
10	Merit List Generation & Export
11	Analytics & Reporting Dashboard





al

of concurrent test-takers during peak

nteractions for question navigation

stem, input validation, and session

ve design (360px+), WCAG-
BN).

seconds with full session recovery on

onent architecture with clean

Requirements

anages the entire system — question
didate records, merit lists, and
all modules.

rs with program selection, reviews
sessment, and views results with

Uses analytics dashboards, pass
ata-driven admission decisions across



ecture

r v7 · Tailwind CSS v4 · Framer Motion ·

React Query (server state) · React Hook

(auto-memoization) · Babel plugin

l-ready) · localStorage persistence ·

d line charts for admin analytics

Architecture

tructure

tions

T
10 min
ware,
OOP

English
10 Qs · 10 min
Grammar, vocabulary

Business
10 Qs · 10 min
Finance, marketing

ionally
ible
verall but

Not Eligible
< 40% overall OR
< 40% in 3+ sections

-2 sections

Assessment

stem

2 **Fullscreen Enforcement**
Exit detection with re-enter prompt, auto-submit at 3 exits

4 **Right-Click Disabled**
Context menu blocked and event logged

6 **DevTools Detection**
Window resize heuristic check every 3 seconds

-question time tracking for admin review

Security

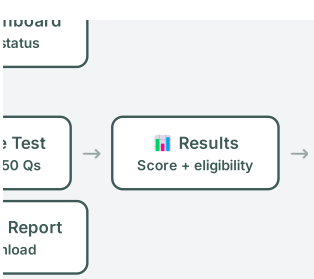
N

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ogram

→

🔐 Login
Credentials

→







stion

· evels · · Section lk CSV

⚙️ Exam Config

Section setup · Time allocation · Passing thresholds · Date windows · Max attempts

t List

ed by
gibility
Top 3
d · CSV

📈 Analytics

Section averages (bar) · Eligibility distribution (pie) · Score trends (line)

Admin

n	Purpose
	Component framework with React Compiler
	Dev server + production bundler
	Utility-first responsive design
	Lightweight state with persistence
	SPA routing with role guards
	Admin analytics visualizations

Client-side report generation

Schema validation + form handling

Tech Stack



ap

ss + PostgreSQL for real data
port.

ased one-time password for secure

+ Bangla (EN/BN) interface for

ent-specific exam configurations and

d hosting with CDN, SSL, and auto-

affic.

Roadmap



k You

Discussion

-Wide Aptitude Assessment
form

y · Daffodil Smart City, Birulia,
Dhaka

